

Function notation



1 $f(7) = -6$ or $f : 7 \rightarrow -6$

2

a 112

b 966

3

a -20

b 4

4

a 12

b -16

5

a 8

b -13

6

a $\frac{35}{4}$

b -7

7

a 68

b -4

8

$\frac{9}{2}$

9

a 3

b $\sqrt{19}$

c $\sqrt{4}$

10

a 0

b 3

c $\frac{91}{25}$

11

a $a^2 + 8a$

b $b^2 + 8b$

12

a -4

b $-\frac{121}{32}$

13

a 0

b 2.67

c 80.99

14

a $\sqrt{128}$

b 12



c $\sqrt{63}$

d $\sqrt{142}$

15

a 12

b $m^2 + 8$

16

a $y = 2 - \frac{x}{3}$

b $f(x) = 2 - \frac{x}{3}$

c $f(3) = 1$

17

a $y = \frac{x}{4} - 2$

b $f : x \longrightarrow \frac{x}{4} - 2$

c $f : 12 \longrightarrow 1$

18

a $y = 3 - x - 6x^2$

b $f(x) = 3 - x - 6x^2$

c -54

19

a $y = 4x^2 + x + 5$

b $f : x \longrightarrow 4x^2 + x + 5$

c 23

20

a $y = \frac{6}{5}x + \frac{7}{5}$

b $f : x \longrightarrow \frac{6}{5}x + \frac{7}{5}$

c 5

21

a 2

b 0

c $x = 1$

22

a 1

b Yes, because every value of x corresponds to only one value of y .

23



a

- Independent variable: Size of pizza
- Dependent variable: Cost of pizza

b

- Independent variable: Time spent studying
- Dependent variable: Mark achieved on a test

c

- Independent variable: Amount borrowed
- Dependent variable: Duration of a loan

d

- Independent variable: Time spent exercising
- Dependent variable: Physical fitness level

24

a

i -48 ii 15 iii -49

b $x = 7.81, x = -7.81$

25

a

 $ak^3 - 3k + 5$

b

 $-ak^3 + 3k + 5$

c

No

d

No

Applications

26 $y = -7, -5$

27

a

320

b

 $x = -4, x = 2, x = -2$

28

a

 $c = -7$

b

 $a = b - 1$

c

 $21 = 4a + b$

d

 $a = 4, b = 5$

29



a $P(x) = -\frac{x^2}{4} + 35x - 410$

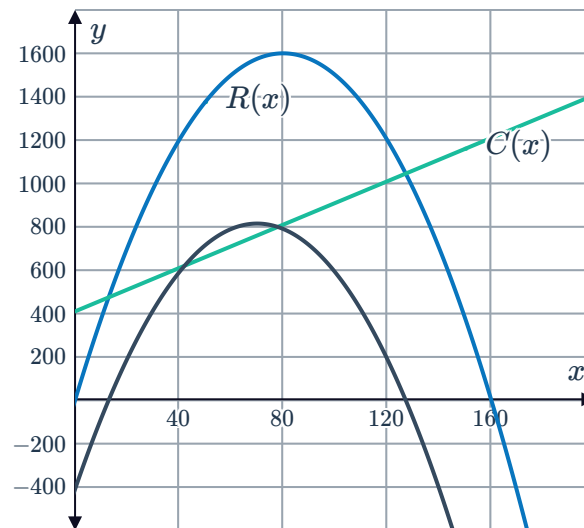
b

i $R(70) = 1575$,

ii $C(70) = 760$

iii $P(70) = 815$

c



30

a $T(x) = x^2 - 182x + 25650$

b

i $S(91) = 308$

ii $U(91) = 17061$

iii $T(91) = 17369$

c

